Expert Systems With Applications 233 (2023) 120925

Contents lists available at ScienceDirect

Expert Systems With Applications

journal homepage: www.elsevier.com/locate/eswa

Deep learning based concrete compressive strength prediction model with hybrid meta-heuristic approach

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ARTICLE INFO

Keywords:
Concrete Compressive Strength
Long Short-Term Memory (LSTM)
Hybrid classifier
Deep Belief Network (DBN)
Rock Hyraxes Updated Sparrow Algorithm (RHUSA)
Fiber Reinforced High Strength Self-Compacting Concrete

ABSTRACT

In concrete design, the compressive strength of the concrete is the critical parameter that defines the quality of concrete. Determination of the compressive strength of concrete by laboratory testing is the conventional way, however, with the help of advancements in technology, the development of an exact predicting technique for the compressive strength of concrete will reduce time as well as cost to a greater extent. There are various types of concrete, as concrete is the most used material and has wide applications in the construction sector and hence huge article is being done to increase the performance of concrete. In this paper, the compressive strength of High-Performance Concrete and Fiber Reinforced High Strength Self-Compacting Concrete is predicted by next three main stages, namely (a) pre-processing, (b) feature extraction, and (c) prediction of concrete compressive strength. Initially, in the data pre-processing phase, duplicate data samples are removed. Subsequently, various features like statistical features, symmetrical uncertainty-based features, higher order statistical features, ReliefF feature, and are derived from the pre-processed data. In the prediction phase of concrete compressive strength, a hybrid algorithm is designed by mingling the concepts of Deep Belief Network and Long Short-Term Memory. Moreover, The Rock Hyraxes Updated Sparrow Algorithm is used to fine-tune the weights of DBN to improve the prediction accuracy of the suggested work. The suggested hybrid optimization method is the hybridization of the standard Sparrow Search Algorithm and Rock Hyraxes Swarm Optimization, respectively. Eventually, the output from the optimized hybrid method will reveal the compression energy of concrete. Ultimately, the efficiency of the suggested method is verified by the comparative estimation. The MAPE of the suggested method is 235.32, which is 30.6%, 21.64%, 20.4%, and 11.4% superior than the CLPSO, ABC, SSA + Hybrid Classifier and RHSO + Hybrid Classifier, respectively.

1. Introduction

In this century, the most commonly preferred construction material is Concrete, which is being the only building material that could be delivered in the plastic state to the construction sites. This unique quality of concrete makes it much more applicable for building materials, as it could be modeled into any form. Concrete is a mixture of Cement, Sand, Coarse Aggregates, and Water. In the modern era of the building sector, taking into account sustainable development is a crucial prerequisite. A computational method called machine learning (ML) uses ongoing learning and evolution to emulate human intelligence and accelerate computation. The very important construction things today is concrete. It is utilized in a variety of applications, including civil engineering, the mechanical industry, shipbuilding, and other projects, because of its integrity, good durability, and affordability (Liu, 2022; Falah et al., 2022; Li et al., 2023; Pakzad et al., 2023). Concrete is utilized to construct a broad range of structures starting from small Bungalows, Buildings, High raised structures, Roads, Bridges, Airports, and Irrigation structures such as Canals, Dams etc. The novel category of high-performance concrete is the SCC, which has been widely accepted for its higher spreading ability and its self-consolidation exhibiting higher constituent separation (Chiew, 2019; Ma et al., 2020; Singh & Khaskil, 2020; Zhang & Zhao, 2017). The Self Compacting Concrete is considered to the special concrete, which is built with a melding of cement, aggregates, steel fibers, and water and different proportions of mineral admixtures (keleş et al., 2018; Ping & Peng, 2011; Abdalla et al., 2011; Wang et al., 2017). These chemical admixtures like the viscosity-

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https://doi.org/10.1016/j.eswa.2023.120925
Received 21 March 2023; Received in revised form 21 June 2023; Accepted 28 June 2023
Available online 3 July 2023
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Nomenclature

Abbreviation	Description
ANN	Artificial Neural Network
DBN	Deep Belief Network
RHSO	Rock Hyraxes Swarm Optimization
OPC	ordinary Portland cement
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
RHUSA	Rock Hyraxes Updated Sparrow Algorithm
SSC	self-compacting concrete
RMSE	Root Mean Square Error
POFA	Palm Oil Fuel Ashes
SEM	scaling electron microscope
RBM	Restricted Boltzmann Machine
SSA	Sparrow Search Algorithm
UPV	ultrasonic pulse velocity

modifying agents and Superplasticiser typically vary the rheological properties of concrete. In addition, the Superplasticiser is used as extra fine material (Saad et al., 2016; Shafiq, 2016; Zhang et al., 2014; Chung et al., 2015).The self-compacting concrete's durability depends on the homogeneous and dense characteristics of this concrete. Recently, the fibers are highly being preferred to improve the concrete's residual properties.

The Compressive Strength (Sinha et al., 2019; Poongodi et al., 2018; Silva et al., 2020; Clemente et al., 2018) of the concrete reflects the quality of the concrete. Typically, on a concrete cube, the standard crushing test is undergone to determine the compressive strength. This testing procedure required engineers to cast small concrete cubes with diverse raw material combinations, and the variations of the strength. Testing a cube requires 28 days of wait time. As a consequence of this, the cost incurred for testing increases and at the similar period, it requires a huge time and labor for various prototype preparation. On the other hand, these techniques are highly prone to human errors, as a minute mistake made during the test can dynamically increase the wait time (Regin et al., 2019; Melo & Banzhaf, 2015; Bigdeli & Barbato, 2017).

Machine learning approaches are preferred to solve the problems of finite accuracy and weak robustness for guessing the compression energy of maximum performance of SCC. On the other hand, conventional ML as well as shallow learning models has exhibited less prediction accuracy. Deep learning techniques are being highly utilized in different engineering fields. Even though the deep learning techniques provide higher exceptional ability in dealing with complex non-linear relationships, the prediction accuracy of the deep learning models is indeed higher than the traditional machine learning models. Optimized deep learning methods can be a one-stop solution to the complex optimization issues that arise in different engineering dimensions.

The major contributions of this paper is:

- Recommends a novel hybrid technique to predict the compression strength of the concrete. The hybrid classifier is the conceptual blending of Deep Belief Network (DBN), and Long Short-Term Memory (LSTM) respectively.
- To increase the prediction accuracy of concrete-compressive strength, the novel proposed Rock Hyraxes Updated Sparrow Algorithm (RHUSA) is used to fine-tune the weights of DBN.

The remaining section of this work is arranged as follows: Section 2describes the related papers in concrete compressive strength prediction. Section 3 and Section 4 tell about the proposed deep learning-based concrete compressive strength prediction method and feature extraction and pre-processing, correspondingly. Additionally, Section 5 describes the information regarding a hybrid classifier for concrete compressive strength prediction. Section 6 addresses the results and discussions of the suggested model. The conclusion of this paper is described in Section 7.

2. 2.Literature review

2.1. Related works

Paji et al. (2021) have investigated the impact of compressive strength by means of assessing them for 28 days with both new and saline water. The compressive strength of the concrete was predicted using two-hybrid AI models: "neuro-swarm and neuro-imperialism". With the intention of acquiring escalated performance, the weight, as well as biases of the ANN, was optimized by using the particle swarm optimization and imperialist competitive classifiers. The evaluation was carried out by using water-to-cement ratio; and diverse content of cement, water rotation time, and magnetic field intensity, all these evaluations have been carried out in the laboratory environment.

Moradi et al. (2021) have estimated the concrete's compressive strength by having the MK with several materials. These evaluations have been carried out by using the accessible experimental outcomes. The outcomes of the analysis show undergone with 28-day compressive strength show that the sensitivity analysis has been influenced mostly by the SiO_2/Al2O3 ratio and the compressive strength.

Zeng et al. (2021) have investigated the air void structural characteristics in terms of the compressive strength at usual air pressure area (P = 1 atm) and minimum air pressure area (0.7 P = 0.7 atm). The air-void micromorphology as well as structure was assessed by means of employing the SEM and hardened air-void structure analyzer.

Jin et al. (2021) have presented a numerical investigation with the aim of investigating the result of the dynamic biaxial compression failure and size of the concrete for the lateral stress ratio and strain rate. In addition, they have introduced a 3D mesoscale modeling approach based on concrete heterogeneity. They have also simulated the specimens having the automatic reaction of cubic concrete with dynamic biaxial compressive conditions and various sizes, and these evaluations were undergone in terms of the strain rates and lateral stress ratios. The outcomes describes that there was an improvement in the strain rate, while the dynamic biaxial compressive strength had slowly poored under the same lateral stress ratio.

Alabi & Mahachi (2020) have assessed the concrete compressive strength during the various curing procedures. The concrete taken for evaluation has been generated at percentage levels 0%, 10%, and 15%, by partially replacing OPC with POFA. The POFA's influence as well as the curing technique's effects was assessed by means of computing the concrete's compressive strength. The outcome from the analysis exhibited that the concrete's compressive strength got increased as the POFA's percentage escalated.

Mahakavi & Chithra (2021) have analyzed the effects of the crimped threads of steels upon the "fiber reinforced self-compacting concrete with RCA and also foundry sand" (FR-SCC). The FR-SCC's automatic materials were boosted when the right proportions of the foundry sand as well as combining fibers with RCA were used. In this study, the authors have evaluated the hardened properties in terms of "compressive, break tensile, flexural forces, and also toughness".

Sliseris & Korjakins (2019) have investigated the concrete's effective compressive strengths as well as the intersection of the exterior column members by means of proposing a new analytical approach. This approach has made the estimation by means of minimizing the concrete slab's strengths. At the column and slab member's intersection, the strain distributions and constraint conditions were derived theoretically by the projected model.

Choi et al. (2018) have deployed a novel simulation architecture for flow modeling during the casting process. The proposed work included

the mathematical modeling of the movement regarding maximum-execution self-compacting fiber-reinforced concrete mortar. This was based on the mortar's deformation gradient and local fiber orientation's computation. They've also projected a deformation gradient to compute the fiber orientation distribution's probability utilizing tracking the motion of the particles. On simulating the Single and three-point concrete casting techniques, it was revealed that there exists a good agreement amongst them.

2.2. Review

In literature, a considerable volume of work has been devoted to predicting the compressive strength of the concrete. These researchers have utilized various ML methods as well as deep learning techniques. Traditionally, experimental designs with experimental data have been highly preferred among researchers for predicting the compression strength of the concrete. Nowadays these techniques are highly improvised with advanced AI-based techniques. It has been widely preferred for pattern recognition and function estimation. Moreover, has the capability of dealing with imprecise or incomplete data. But, here the prediction accuracy is lower. On the other hand, the classical regression methods exhibited higher skewness, which ought to be reduced. The training errors are higher with most traditional neural network models, which is a major hurdle. Therefore, the deep learning models can be preferred, as an alternative.

3. Proposed deep learning based concrete compressive strength prediction method

Concrete construction are frequently in a complicated stress condition for a more period in realistic construction. In this research work, we are going to guess the compression strength of High-Performance Concrete and Fiber Reinforced High Strength Self-Compacting Concrete by next three major phases, namely: (a) pre-processing, (c) feature extraction, and (d) prediction of concrete compressive strength. Fig. 1 illustrates the architecture of the suggested method. Initially, the data has been collected from six different concrete strength grades: M50, M60, M70, M80, M90, and M100. In the data pre-processing stage, the duplicate information available within the collected information were eliminated. Subsequently, multi-features like higher order statistical features (variance and skewness), statistical features (mean, median, min–max, standard deviation), ReliefF feature, and symmetrical uncertainty-based features are derived from the pre-processed data. In the prediction of the concrete compression strength stage, a hybrid classifier is structured by mingling the models of DBN and LSTM. Moreover, the novel hybridization method is useful to tuning the weights of DBN to improve the prediction accuracy of the adopted model. The proposed hybrid optimization model is the hybridization of standard SSA (Xue & Shen, 2020) and RHSO (Al-Khateeb et al., 2021), respectively. Eventually, the output from the hybrid model will shows the compressive strength of concrete.

4. Pre-Processing and feature extraction

4.1. Pre-Processing

The collected input data D^{inp} (M50, M60, M70, M80, M90, and M100) is subjected to pre-processing phase, wherein the data duplicate removal takes place. The pre-processed data is denoted as D^{pre}. From this D^{pre}, the most renowned multi-features will be extracted.

4.2. Feature extraction

The multi-features like higher order statistical features (variance and skewness), statistical features (mean, median, min–max, standard deviation), ReliefF feature, and symmetrical uncertainty have been extracted from D^{pre}.

4.2.1. Statistical features

Mean: The mean f^{mean} or average of D_i^{pre} is computed by finding the addition of observations to its number of observations N. Mathematically, median is depicted in Eq. (1)

$$\mu = \frac{\sum_{i=1}^{N} D_i^{pre}}{N} \tag{1}$$

In MATLAB, the mean has been computed as per Eq. (2).

$$f^{mean} = mean(D_i^{pre}) \tag{2}$$

Median: The middle value within D_i^{pre} is said to be the median. If D_i^{pre} is a set of even observations, then the average is computed between the two middle values, and this is said to be the median value. The media feature is more significant in dividing the data into two equal halves.

In MATLAB, the median f^{median} has been computed using the median () function as shown in Eq. (3).

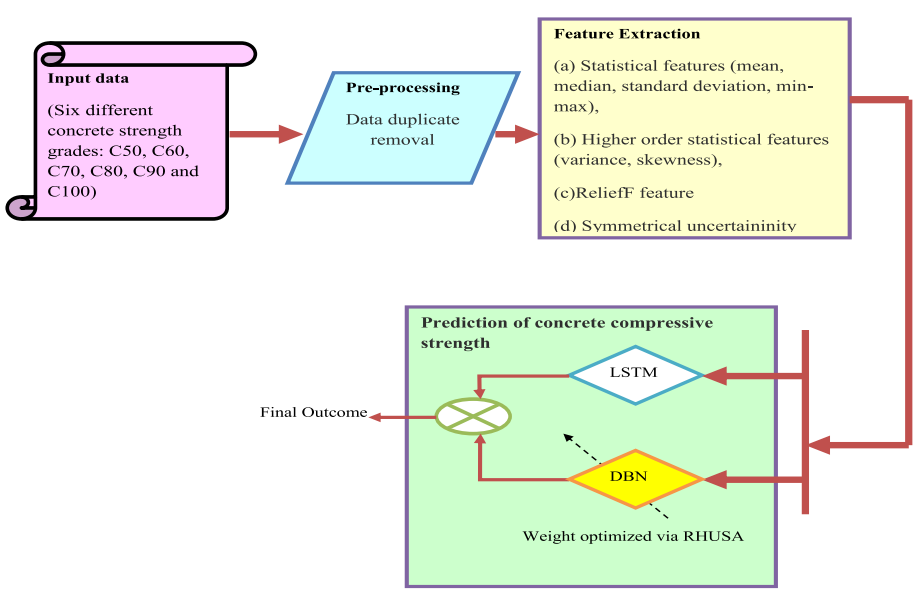


Fig. 1. Framework of the proposed model.

$$f^{median} = median(D_i^{pre}) \tag{3}$$

Standard Deviation: The SD is computed to find the closeness of the entire dataset to the average value. The dataset having the lowest SD portrays that the data within the datasets are closely packed and precise, while the dataset with the highest SD says that the data within the dataset are distributed over a extensive range of values. Mathematically, the mean can be given as per Eq. (4).

$$o = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(D_i^{pre} - \mu)^2} \tag{4}$$

In MATLAB, the SD has been computed using the std() function as per Eq. (5).

$$f^{std} = std(D_i^{pre}) \tag{5}$$

Min-Max: The min is the minimal rate in D_i^{pre}, while max is the maximal rate within D_i^{pre}. In MATLAB, the min $f^{\min}$ and max $f^{\max}$ values are computed using the min() and max() function. Mathematically, the min and max can be provided as per Eq. (6) and Eq. (7), respectively.

$$f^{\min} = \min(D_i^{pre}) \tag{6}$$

$$f^{\max} = \max(D_i^{pre}) \tag{7}$$

The extracted statistical features (mean, median, standard deviation, min–max) is denoted as F^{stat}=f^{mean}+f^{median}+f^{std}+$f^{\min}$ + $f^{\max}$.

4.2.2. Higher order statistical features

Variance: It is the measure of the squared differences of the data from the median. The variance provides information regarding the spreading of all the data points corresponding to D_i^{pre}. The extracted variance feature is denoted as f^{var}. Mathematically, the variance can be given as per Eq. (8).

$$f^{var} = var(D_i^{pre}) \tag{8}$$

Skewness: The skewness is the calculate of the cubed differences of the data D_i^{pre} from the mean μ.the skewness measure provides information regarding the right (negative skew) or left skew (positive skew) of the data. The curve for the right tail is longer in positive skew, whereas, the curve for the left tail seems to be longer in negative skew. f^{skew} refers the extracted skewness feature.

$$f^{skew} = skewness(D_i^{pre}) \tag{9}$$

F^{high}=f^{skew}+f^{var} indicates the higher-order statistical features.

Symmetric Uncertainty: It is utilized for finding the non-linear relationship amongst the variables and class labels and between the variables of D_i^{pre}. The SU is formulated based on the information gain as well as entropy. This SU aids in computing the fitness of the features. The features with high SU will be selected for training the hybrid classifier. Mathematically, the SU can be given as per Eq. (10).

$$SU(i_1, i_2) = \frac{2 \times MI(i_1, i_2)}{H(i_1) + H(i_2)} \tag{10}$$

Here, $MI(i_1, i_2)$ denotes the mutation information of two random features i_1,i_2. In addition, $H(i_1)$ and $H(i_2)$ are the conditional entropies of discrete random variables i_1, i_2. f^{SU} indicates the extracted SU feature.

ReliefF feature: For each of the feature D_i^{pre}, the proximity statistics is computed using the relief feature. This ReliefF feature estimates the proxy statistic for every feature. These feature statistics are denoted as the feature weights or the 'feature score'. $f^{ReliefF}$ indicates the extracted ReliefF feature.

F indicates extracted overall features.

$$F = f^{ReliefF} + f^{skew} + F^{high} + F^{stat} \tag{11}$$

5. Hybrid classifier for concrete compressive strength prediction

In this paper, a hybrid classifier has been designed to acquire the final predicted values. The suggested hybrid model has been evaluated by mingling the DBN and LSTM, correspondingly.The derived features F is used to train the both LSTM and DBN parallelly. The output from DBN and LSTM is finally combined by computing the average. This averaged value is the final predicted compressive strength of the concrete value.

5.1. LSTM

LSTM is a deep learning framework that was related on the artificial recurrent neural network (RANN (Moradi et al., 2021)). The LSTM is a sort of RANN (Moradi et al., 2021) with respect of memory, but functions superior than it. The LSTM outperforms the RANN (Moradi et al., 2021) in memorizing specific patterns. LSTM encapsulates a huge count of hidden layers. The derived features F is used to train the LSTM. The input, forget, and output gates are referred as gates Inp, for, out.

1. FORGET Gate: This gate provides information regarding the data that need to be retained and data that must be deleted. The data acquired from the preceding hidden stage hid_{t-1} is given as the input to the current state x_t. Mathematically, the forget gate is expressed in Eq. (12).

$$for = \sigma(x_t.Q_{for} + hid_{t-1}.W_{for}).inp = \sigma(x_t.Q_{for} + hid_{t-1}.W_{for}) \tag{12}$$

Here, W_{for} is the weight of the forget gate

2. INPUT Gate: The cell states are modified by the input gate to explore the relevant data from F.The input gate such as forget gate provides the findings of important data besides the memory storage of applicable data. Along the sigmoid and tanh functions, the inputs x_t and hid_{t-1} are passed. Mathematically, the input gate is expressed in Eq. (13).

$$Inp = \sigma(hid_{t-1}.W_{inp} + x_t.Q_{inp}).inp = \sigma(hid_{t-1}.W_{inp} + x_t.Q_{inp}) \tag{13}$$

Here, W_{inp} is the weight of the input gate

3. Cell State: With all the gathered information, the computation of the new cell state takes place. Initially, the state is increased by the forget gate's output. If the resultant of the multiplication is 0(zero), then the cell state is dropped.

4. OUTPUT Gate: it is the final state. Here, with hid_{t-1} and x_t the sigmoid process is defined. Then, the information that needs to be stored within the hidden layer is decided using sending the newly implemented cell state via the tanh function, then it is multiplied with the sigmoid output. Mathematically, the output gate is expressed in Eq. (14).

$$out = \sigma(hid_{t-1}.W_{out} + x_t.Q_{out}).inp = \sigma(hid_{t-1}.W_{outr} + x_t.Q_{out}) \tag{14}$$

Here, W_{out} is the weight of the output gate

5. The unit's inner memory is denoted as C_t. A combination of the old C_{t-1} memory, the hidden state g, and the forget gate, the presently calculated, multiplied by the input gate. Mathematically, C_t is shown in Eq. (15).

$$C_t = for_t + C_{t-1} + inp_t{}^*g \tag{15}$$

5.2. DBN

The DBN encapsulates multiple layers, in which there are visual and invisible neurons. The visual neurons are entirely combined with the invisible neurons. In general, the imaginary neuron's result ($output$) from the Boltzmann networks is presumption in nature. Numerically, $output$ (binary form) is given as per Eq. (16), in which $prob(\xi)$ is the possibility function and $Temp$ is the pseudo-temperature. For Boltzmann machine, configured energy state (q) and the energy ($Energy$) is described as per Eq. (18), wherein the weight function between neurons A and B is denoted as $wt_{A,B}$. Besides, the biases are depicted by the symbol ψ_A. Further, in the RBM, the acquired new energy function from the visual and invisible neurons ($visible, hiddle$) is shown in Eq. (20). In

unsupervised way, the RBM has been trained and accomplished.

$$output = \left\{ \begin{array}{ll} 0 & with\,probability\,of\ 1 - prob(\xi) \\ 1 & with\,probability\ of\quad prob(\xi) \end{array} \right\} \tag{16}$$

In which,

$$prob(\xi) = \frac{1}{1 + e^{\frac{-\zeta}{Temp}}} \tag{17}$$

$$Energy(q) = -\sum_{A<B} q_A\, q_B\, wt_{A,B} - \sum_{A} \psi_A\, q_A \tag{18}$$

$$Energy(q) = -\sum_{A<B} q_A\, q_B\, wt_{A,B} - \sum_{A} \psi_A\, q_A \tag{19}$$

$$Energy\left(\overrightarrow{visible}, \overrightarrow{hidden}\right) = \sum_{(A,B)} wt_{A,B}\, visible_A.hidden_B - \\ -\sum_{A} visible_A P_A - \sum_{B} hidden_B Q_B \tag{20}$$

$$Energy\left(visible_A, \overrightarrow{hidden}\right) = \sum_{B} wt_{A,B}.hidden_B + P_A \tag{21}$$

$$Energy\left(\overrightarrow{visible}, hidden_B\right) = \sum_{A} wt_{A,B}.visible_A + Q_B \tag{22}$$

Within the training set *Train*, $\overrightarrow{visible}$ is represented by row matrix. Moreover,$Weight_m$(weight assignment) is used for accomplishing the maximal probability. The energy function is used to define a probability that is allocated to every hidden and visible neuron pair $(visible, hidden)$ in Eq. (23).

$$Weight_m = \max_{Weight} \prod_{\overrightarrow{cisible} \in Train} prob\left(\overrightarrow{visible}\right) \tag{23}$$

The partition function Z is mathematically computed as per Eq. (25).

$$\frac{1}{Z} e^{-E^{Fun}\left(\overrightarrow{visible}, \overrightarrow{hidden}\right)} = prob\left(\overrightarrow{visible}, \overrightarrow{hidden}\right) \tag{24}$$

$$\sum_{\overrightarrow{visible}, \overrightarrow{hidden}} e^{-E^{Fun}\left(\overrightarrow{visible}, \overrightarrow{hidden}\right)} = Z \tag{25}$$

During the compressive strength prediction process, the error (Mean Square Error (MSE)) that takes place in prediction ought to be reduced. This in turn paves the way for the increase in prediction accuracy. Therefore, the objective function of this model has been set as the minimization of the error function of DBN. The main contribution of this paper is described in Eq. (26). Mean Square Error (MSE) is defined as the difference between predicted outcomes *Pre* and actual outcomes *Act*.

$$Obj = Min(MSE) \tag{26}$$

$$MSE = Pre - Act \tag{27}$$

Further, the Contrastive Divergence (CD) has been introduced with the objective of initializing the visual states in a systematic way. The procedures under the CD are described here:

1. Onto visible neurons, the *visible* is claimed
2. Find the probability of hidden neurons $prob_{hidden} = \sigma(visible.\,Weight)$
3. Calculate Positive gradient φ^+ for $\overrightarrow{visible}$ and $\overrightarrow{Weight}$.
4. Then, sample the reformation of the visible state $visible'$ from *hidden*.
5. Calculate Negative gradient $\varphi^- = visible'.\,hidden'^T$
6. Weight update as $\Delta Weight = \delta(\varphi^+ - \varphi^-)$
7. The current weight updating is as per Eq. (28). This *wt* is fine-tuned using the proposed RHUSA model.

$$wt_{visible,hidden} = \Delta wt_{visible,hidden} + wt_{visible,hidden} \tag{28}$$

5.3. Rock Hyraxes Updated Sparrow Algorithm (RHUSA).

The proposed optimization method is the theoritical improvement of standard RHSO and SSA. The RHSO has been improved depends on the collective behavior (in terms of seeking food and eating) of the rock hyraxes swarms. The RHSO algorithm is highly effective in solving real optimization issues within the search space, and it could exhibit higher performance in terms of convergence. In addition, the SSA was developed with the motivation recorded from the anti-predation behaviors of sparrows, foraging, and group wisdom. The SSA mode is good for resolving difficult engineering problems without getting caught in the local optima. In the literature, it is defined that the hybridization of two optimization algorithms increases the convergence of the solutions (Shailaja & Guru Rao, 2019). In this research work, SSA is implied within the RHOS model and hence named as Rock Hyraxes Updated Sparrow Algorithm (RHUSA). The weight Wt_N of DBN is the input (solution) to the RHUSA method. Fig. 2. indicates the solution encoding.

The procedures comes under the RHUSA technique are described here:

Step 1: Assign the population *Pop* of *M* search agents. The maximal iteration is set as $\max^{itr}$ and the current iteration is *itr*.

Step 2: Compute the fitness of the search agent using Eq. (26).

Step 3: The best search agent is allocated as the leader *L*.

Step 4: fix$itr = 1$

Step 5: While$\max^{itr} < itr$do

(a)Using Eq. (29), update the position of the leader.

$$L = X(L_{pos}, J) \times rand_1[0,1] \tag{29}$$

Here, *L* is the leader, and its old position is denoted as L_{pos}. In addition, $rand_1[0,1]$ is the random number produced within the range 0 to 1. The existing position of the leader is denoted as the notation *X* and *j* is the dimensions.

(b)Using the presently evaluated term shown in Eq. (2), update the position of the search agent. Our objective consists in this stage. Here, the RHSO's leadership position is updated using the SSA algorithm. The presently evaluated term is shown in Eq. (30).

$$\left(X(i,j) - (Circ^*.\left\{X_{best}(i,j) + \beta\left|X(i,j) - X_{best}(i,j\right|\right\} + L)\right) = X(i,j) \tag{30}$$

Here, the global best position of the leader is referred as $X_{best}(i,j)$ and *Circ* denotes the circular motion that is computed as per Eq. (31)

$$Circ = sqrt(N_1^2 + N_2^2) \tag{31}$$

$$N_1 = rand_2[0,1]^*\mathrm{Cos}(Ang) \tag{32}$$

$$N_2 = rand_2[0,1]^*\mathrm{Sin}(Ang) \tag{33}$$

Here, $rand_2[0,1]$ is a random number produced within the range 0 to 1, and *Ang* is also a random number produced within the range 0 to 360. The updating *Ang* takes place across every generation, and this is related on the upper *U* also with lower *L* bounds of the variables.

(c) Compute the fitness of the search agent using Eq. (26).

(d) The best search agent is allocated as the leader *L*.

(e) Using Eq. (34) and Eq. (35), the angle is updated.

Ang=*Ang*+*Dalta* (34)

Dalta=*random*(L, U) (35)

(f) *itr*=*itr* + 1

Step 6: End While

Step 7: return leader *L* (best solution)

6. Result and discussion

6.1. Simulation setup

The adopted method has been executed in MATLAB. The evaluation has been carried out for the six test cases: M50, M60, M70, M80, M90,

Fig. 2. Solution Encoding.

and M100. Dataset 1 and dataset 2 are considered in this work. The dataset 1 was downloaded from https://zenodo.org/record/4620872 and the dataset 2 was installed from https://www.kaggle.com/datasets/elikplim/concrete-compressive-strength-data-set. A total of 600 data have been used (Vakhshouri et al., 2018). Under For test case 1 total 100 data are chosen, if learning rate is 0.7 the training set contain 30 data and the testing set contains 70 data. For test case 2, total 98 data are chosen, if learning rate is 0.7 training set uses the 29 data and testing test uses the 69 data. For test case 4, total 98 data are considered, if learning rate is 0.7 training set uses the 30 data and testing set uses the 70 data. For test case 5, total 98 data are considered, if learning rate is 0.7 training set uses the 29 data and testing set uses 69 data. For test case 6, total 96 data are considered, if learning rate is 0.7 training set uses the 29 data and testing uses the 67 data.For dataset 2 totally 994 images are used. If learning is 0.7, training uses the 298 data and testing uses the 694 data.They include the information: R and Vp (km/s) and output value Actual compressive strength (Mpa). Vp(km/s) stands for ultrasonic pulse velocity (UPV). All these evaluations have been performed in terms of convergence analysis, MAE, MAPE, MSE, and RMSE correspondingly. With the actually computed values, the predicted values acquired with the existing models like CLPSO (Li et al., 2019), ABC (keleş et al., 2018), SSA, and RHSO are compared, and the results acquired are discussed.

The MAE "measures the average magnitude of the errors in a set of forecasts, without considering their direction". The MAE is calculated using Eq. (36). Here, N denotes the count of observations.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\frac{|Act - Pre|}{N} \tag{36}$$

Where, the actual and the predicted compressive strength of the concrete is referred as *Act* and *Pre*, respectively.

"The MAPE, also known as MAPD, is a measure of prediction accuracy in statistics". It is geometrically described as per Eq. (37).

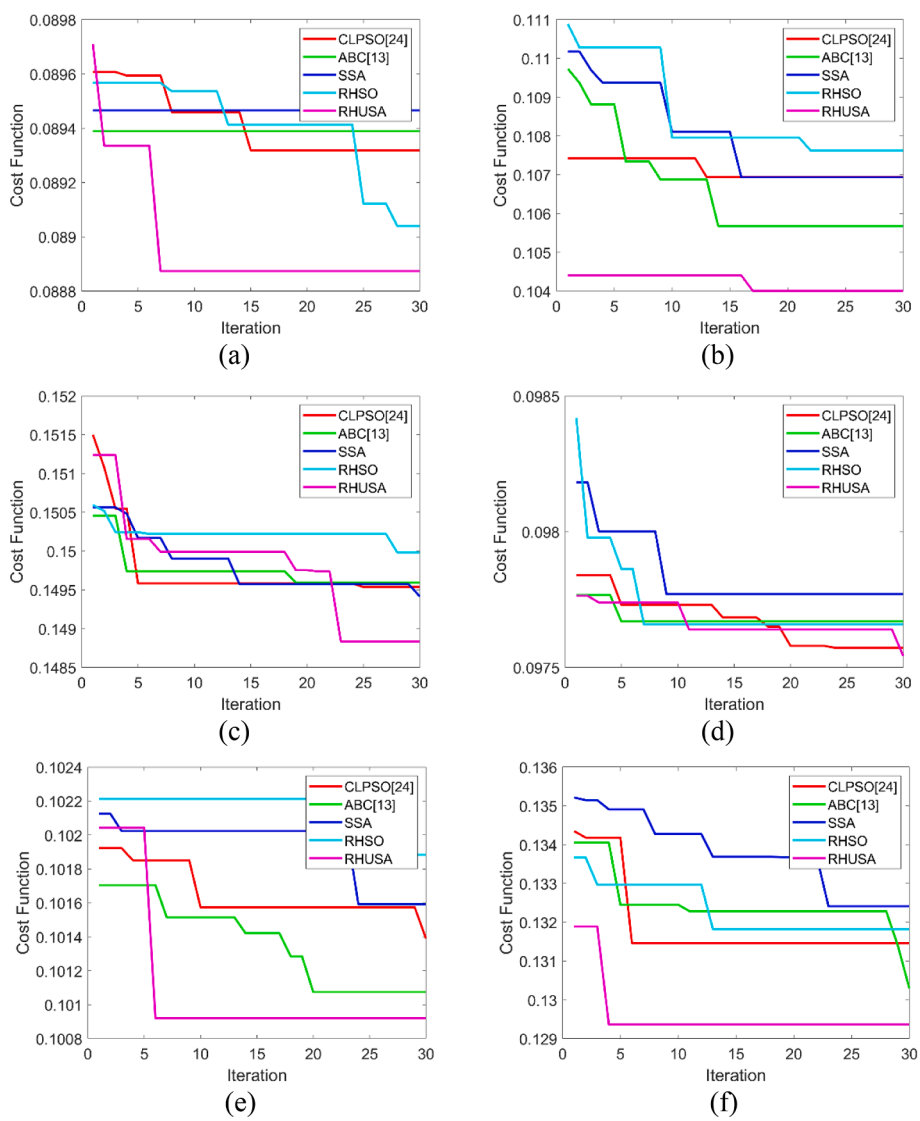


Fig. 3. Convergence Analysis of proposed work for (a)M50, (b) M60, (c) M70, (d) M80, (e) M90 and (f) M100.

$$MAPE = \frac{1}{N}\sum_{i=1}^{N}\left|\frac{Act - Pre}{Act}\right| \tag{37}$$

"The RMSE is a frequently used measure of the differences between values (sample or population values) predicted by a model or an estimator and the values observed". RMSE can be calculated geometrically per Eq. (38).

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Act - Pre)^2} \tag{38}$$

6.2. Convergence analysis for dataset 1

Theoretically, it is defined that the hybridization concepts are improved by the convergence speed of the solutions. However, it is most crucial to prove that the convergence speed of the suggested method is higher than the existing CLPSO (Li et al., 2019), ABC (keleş et al., 2018), SSA, and RHSO, respectively. In this paper, the convergence analysis is performed for 6 test cases: M50, M60, M70, M80, M90, and M100; and the corresponding outputs collected are depicted in Fig. 3. On observing the outcomes visually, the adopted method has indeed achieved the objective function (minimization of error). In the case of M50 test case, the cost function of the suggested besides the conventional method is maximum at the 0 to 5 iterations. After the 6th iteration, the cost function of the adopted method lowered and achieved the least rate of 0.08882. At the 30threpetition, the cost function of the suggested method is 0.67%, 0.78%, 0.90%, and 0.22% better than the convergence of CLPSO (Li et al., 2019), ABC (keleş et al., 2018), SSA and RHSO, correspondingly. Additionally, for M60 the cost function of the adopted method is minimum for every comparison in the iteration count. In the case of M70, the cost function of the suggested model was maximum till the 23rd iteration, and then there was a constant fall in the cost function. Finally, at the maximum iteration count (i.e. 30th iteration), the adopted method has achieved the minimum cost function that all the traditional methods. This portrays that the proposed work could function significantly even under the highest iteration count. Moreover, under M80, the cost function of the suggested method has achieved the lowest rate of as0.09752 at the 30th iteration. In addition, the cost function of the suggested work for the M90 method is maximum between the ranges 0 to 5th iteration count. Then, beyond this range, the cost function of the suggested method had carried out a sudden fall in the cost function, and achieved the least value between the iteration counts 6 to 30. At the 30th iteration count, the cost function of the suggested model is 0.59%, 0.195, 0.98%, and 1.365 better than the convergence of CLPSO (Li et al.,

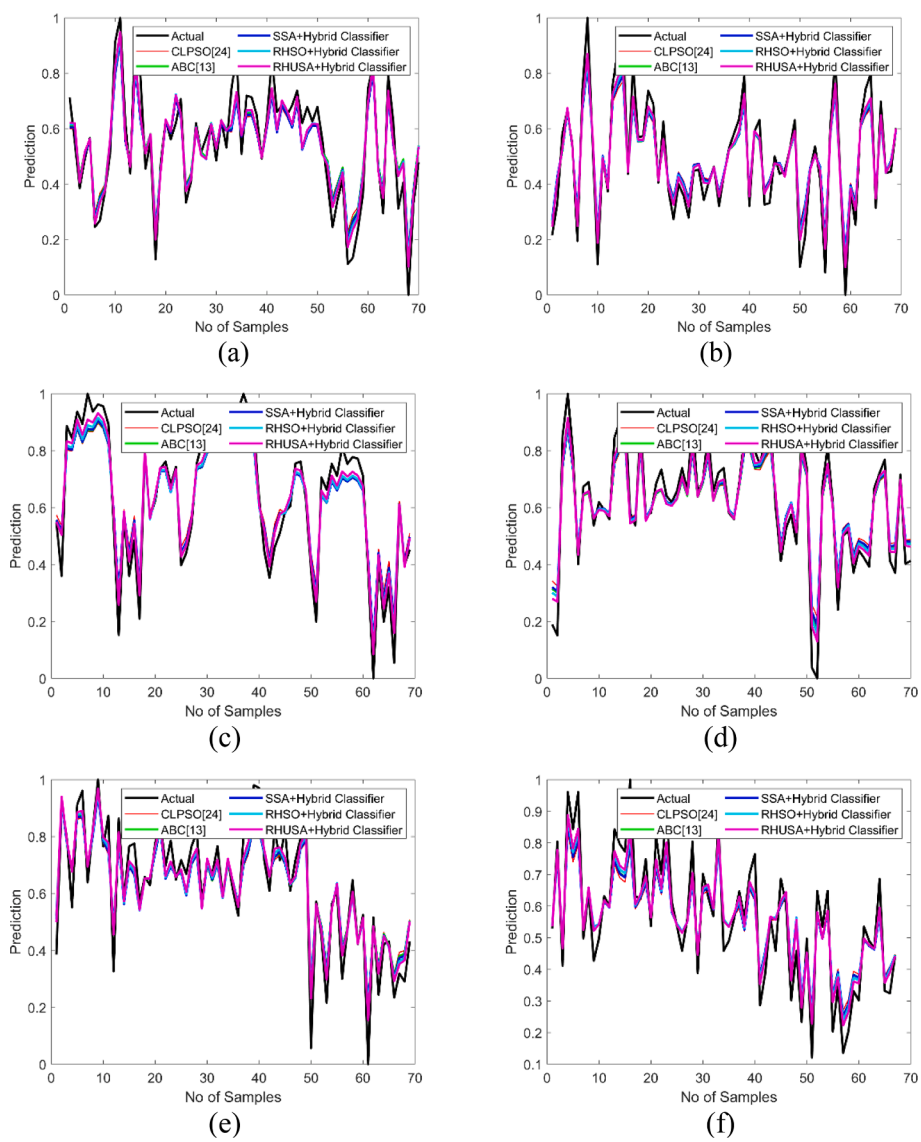


Fig. 4. Algorithmic performance analysis of proposed work for (a) M50, (b) M60, (c) M70, (d) M80, (e) M90 and (f) M100.

2019), ABC (keleş et al., 2018), SSA and RHSO, respectively. Moreover, the cost function of the adopted method is lower for every comparison in the iteration count for M100, and here the suggested method has achieved the minimum cost function as 0.1292 at the 30th iteration count. Thus, from the estimation, it's defined that the suggested method is much more appropriate for predicting the strength of the concrete, as it has achieved the objective of minimizing the errors, and this ultimately says that the suggested model is higher suggested to predict it.

6.3. Performance evaluation for dataset 1: predicted vs actual

Fig. 4 and Fig. 5 shows the predicted as well as actual output values in terms of different algorithmic and classifiers correspondingly. To validate the efficiency of the projected model for automatically predicting the compression strength with higher prediction accuracy, an evaluation has been undergone. The predicted and actual outcomes are compared for the models like CLPSO (Li et al., 2019), ABC (keles et al., 2018), SSA, RHSO, and RHUSA. From the results, it is clear that the guessed values are highly correlated with the usual values. Therefore, it is clear that the prediction error is lower with the proposed RHUSA, and hence it can be highly applicable to predict the strength of the concrete.

6.4. Algorithmic Evaluation

The suggested method has been evaluated algorithmically over the traditional optimization models like CLPSO (Li et al., 2019), ABC (keles et al., 2018), SSA + Hybrid Classifier, RHSO + Hybrid Classifier, correspondingly. The outcomes acquired with M50, M60, M70, M80, M90, and M100 are given in Tables 1–6, respectively. On noticing the outcomes, the suggested method has attained the lowest rate in the case of MAE, MSE, RMSE, and MAPE, correspondingly. This enhancement is owing to the development of the new RHUSA technique for optimizing the weight of DBN. As a consequence of this, the error function of DBN

Table 1
Algorithmic Evaluation of RHUSA + Hybrid Classifier for M50.

Measures	CLPSO(Li et al., 2019)	ABC(keleş et al., 2018)	Hybrid Classifier + SSA	Hybrid Classifier + RHSO	Hybrid Classifier + RHUSA
MAE	0.0684	0.0618	0.0617	0.0562	0.0521
MSE	0.0067	0.0055	0.0055	0.0046	0.0040
RMSE	0.0817	0.0743	0.0742	0.0678	0.0629
MAPE	253.6700	220.1000	211.9800	185.7200	155.8400

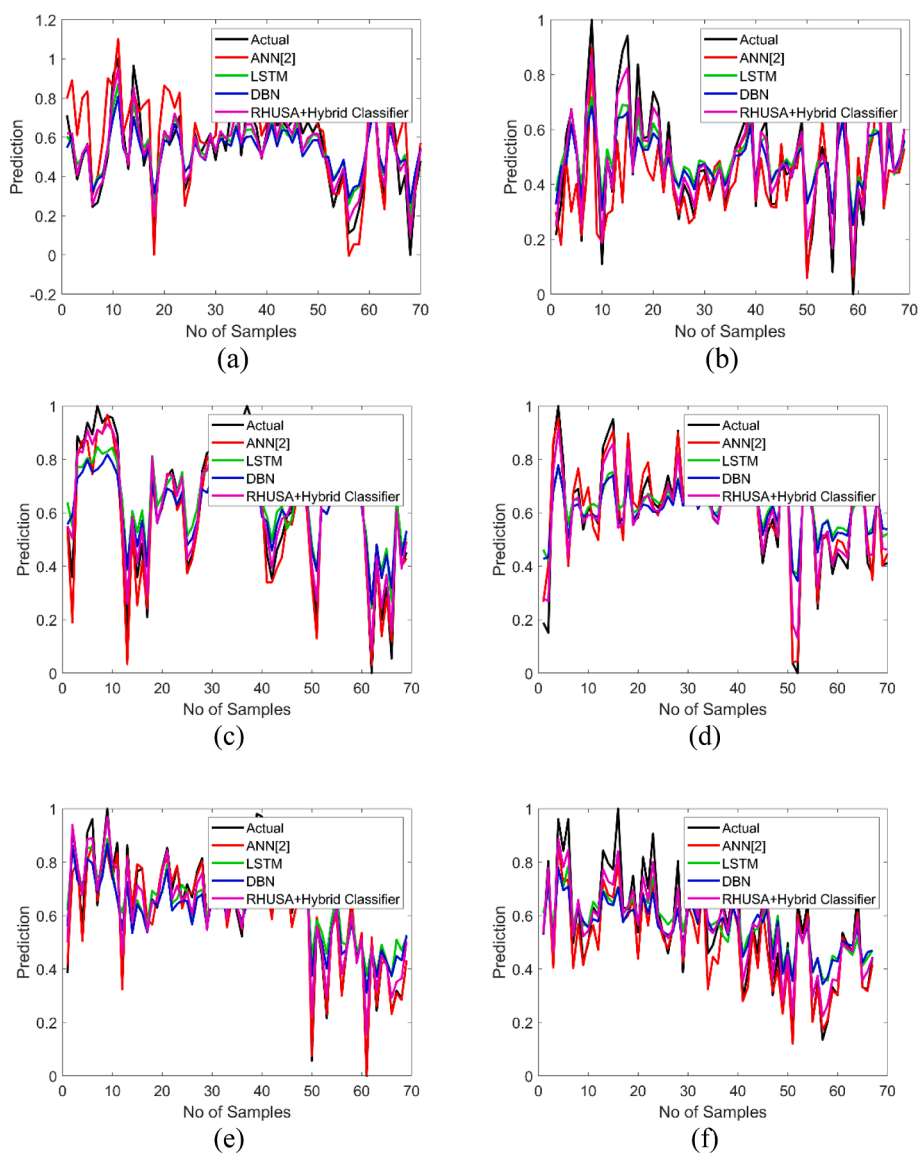


Fig. 5. Classifier performance Analysis of proposed work for (a) M50, (b) M60, (c) M70, (d) M80, (e) M90 and (f) M100.

Table 2
Algorithmic Evaluation of RHUSA + Hybrid Classifier for M60.

Measures	CLPSO(Li et al., 2019)	ABC(keleş et al., 2018)	Hybrid Classifier + SSA	Hybrid Classifier + RHSO	Hybrid Classifier + RHUSA
MAE	0.0606	0.0523	0.0535	0.0468	0.0406
MSE	0.0060	0.0046	0.0047	0.0037	0.0028
RMSE	0.0774	0.0678	0.0689	0.0605	0.0525
MAPE	253.9400	213.5900	214.8700	183.2500	154.7300

Table 3
Algorithmic Evaluation of RHUSA + Hybrid Classifier for M70.

Measures	CLPSO(Li et al., 2019)	ABC(keles et al., 2018)	Hybrid Classifier + SSA	Hybrid Classifier + RHSO	Hybrid Classifier + RHUSA
MAE	0.0692	0.0595	0.0599	0.0519	0.0435
MSE	0.0065	0.0049	0.0049	0.0037	0.0027
RMSE	0.0806	0.0698	0.0701	0.0611	0.0519
MAPE	242.1900	190.1000	192.4200	158.1400	132.5900

Table 4
Algorithmic Evaluation of RHUSA + Hybrid Classifier for M80.

Measures	CLPSO(Li et al., 2019)	ABC (keles et al., 2018)	Hybrid Classifier + SSA	Hybrid Classifier + RHSO	Hybrid Classifier + RHUSA
MAE	0.060	0.053	0.053	0.047	0.041
MSE	0.006	0.004	0.005	0.003	0.003
RMSE	0.077	0.067	0.067	0.059	0.051
MAPE	326.240	272.880	280.920	238.850	196.970

Table 5
Algorithmic Evaluation of RHUSA + Hybrid Classifier for M90.

Measures	CLPSO(Li et al., 2019)	ABC(keleş et al., 2018)	Hybrid Classifier + SSA	Hybrid Classifier + RHSO	Hybrid Classifier + RHUSA
MAE	0.066	0.060	0.060	0.055	0.049
MSE	0.007	0.006	0.006	0.005	0.004
RMSE	0.085	0.077	0.077	0.070	0.062
MAPE	339.310	300.310	295.710	265.840	235.320

got minimized. For M50, the MAE of the suggested model is 0.052132, which is 23.7%, 15.7%, 15.4%, and 7.1% superior than the CLPSO(Li et al., 2019), ABC (keles et al., 2018), SSA + Hybrid Classifier, RHSO + Hybrid Classifier, correspondingly. Additionally, the RMSE of the suggested model is 0.062852, which is 23.08%, 15.3%, 15.3%, and 7.35% better than the CLPSO(Li et al., 2019), ABC (keleş et al., 2018), Hybrid Classifier + SSA, Hybrid Classifier + RHSO, correspondingly. Furthermore, in the case of M60, the MSE of the suggested method is 0.0027602, which is 53.9%, 39.3%, 41.7%, and 24.6% better than the CLPSO(Li et al., 2019), ABC (keleş et al., 2018), SSA + Hybrid Classifier, RHSO + Hybrid Classifier, accordingly. Besides, for M60, the MAPE of the suggested method is 154.73, which is the minimum rate when compared to CLPSO (Li et al., 2019) = 253.94, ABC (keles et al., 2018) = 213.59, SSA + Hybrid Classifier = 214.87, RHSO + Hybrid Classifier = 183.25. On the other hand, for M70, the MAE of the suggested model is 0.043516, which is 37.08%, 26.8%, 27.35%, and 16.17% better than the CLPSO (Li et al., 2019), ABC (keles et al., 2018), SSA + Hybrid Classifier and RHSO + Hybrid Classifier, accordingly. Additionally, the MAPE of the suggested model for M80 is 196.97, which is the least rate of CLPSO (Li et al., 2019), ABC (keleş et al., 2018), SSA + Hybrid Classifier, RHSO + Hybrid Classifier, correspondingly. This is owing to the introduction of the weight-optimized DBN. Additionally, the suggested model has attained the minimum MAE, MSE, RMSE, and MAPE in the case of both M90 and M100 also. Under M90, the MAE of the suggested model is 0.048849, which is 25.5%, 18.14%, 19.08%, and 10.6% better than the CLPSO(Li et al., 2019), ABC (keleş et al., 2018), Hybrid Classifier + SSA and Hybridclassifier + RHSO, accordingly. Moreover, under M90, the MAPE of the suggested method is 235.32, which is 30.6%, 21.64%, 20.4%, and 11.4% better than the CLPSO (Li et al., 2019), ABC (keleş et al., 2018), Hybrid Classifier + SSA and Hybrid-Classifier + RHSO, correspondingly. Under M100, the suggested method has recorded the least MAE, MSE,RMSE, and MAPE rates. The MAE of the suggested k under the M100 case is 0.05137, which is 26.9%, 19.06%, 19.6%, and 11.1% better than the CLPSO (Li et al., 2019), ABC (keleş et al., 2018), Hybrid Classifier + SSA and Hybrid + RHSO Classifier, correspondingly. Hence, from the estimation, it is clear that the suggested model has attained the minimum error metrics.

6.5. Classifier performance

The performance of the suggested method is evaluated over the traditional classifiers like DBN, ANN (Moradi et al., 2021), and LSTM, respectively. The outcomes acquired with M50, M60, M70, M80, M90, and M100 are given in Tables 7–12, respectively. On noticing the results, the suggested method has attained the minimum rate in terms of MAE, MSE, RMSE, and MAPE, accordingly. This is as the result of the development of the RHUSA model for optimizing the weight of DBN. The MAE of the suggested model for M50 is 0.056174, which is 39.8%, 59.6%, and 28.3% better than DBN, ANN (Moradi et al., 2021), and LSTM, accordingly. The MAPE of the adopted method is 185.72 for M50, which is 53.75%, 31.7%, and 41.1% increased over the conventional models such as DBN, ANN (Moradi et al., 2021), and LSTM, respectively. Moreover, under the M60 category, the RMSE of the suggested model is 0.052537, which is 57.3% improved over the DBN, 63.4% improved over the ANN (Moradi et al., 2021) and 55.3% improved over the LSTM. Moreover, an improvement of 77.8%, 12.1%, and 72.7% has been stored by the adopted model over the existing DBN, ANN (Moradi et al., 2021), and LSTM in terms of MSE under M80. Moreover, the MAE of the projected work is 0.046649 for M90, which is 51.6%, 27.8%, and 53.01% improved over DBN, ANN (Moradi et al., 2021), and LSTM, correspondingly. Additionally, the MAE, MSE, RMSE, and MAPE of the suggested method are minimum than the traditional methods in the case of

Table 7
Classifier Performance Evaluation of RHUSA + Hybrid Classifier for M50.

Measures	DBN	ANN (Moradi et al., 2021)	LSTM	RHUSA+Hybrid Classifier
MAE	0.0935	0.1392	0.0784	0.0562
MSE	0.0128	0.0284	0.0089	0.0046
RMSE	0.1131	0.1684	0.0941	0.0678
MAPE	401.8600	272.2900	315.7800	185.7200

Table 6
Algorithmic Evaluation of RHUSA + Hybrid Classifier for M100.

Measures	CLPSO(Li et al., 2019)	ABC(keleş et al., 2018)	SSA+Hybrid Classifier	RHSO+Hybrid Classifier	RHUSA+Hybrid Classifier
MAE	0.070	0.063	0.064	0.058	0.051
MSE	0.007	0.006	0.006	0.005	0.004
RMSE	0.085	0.076	0.077	0.069	0.062
MAPE	16.498	14.844	14.903	13.467	11.814

Table 8
Classifier Performance Evaluation of RHUSA + Hybrid Classifier for M60.

Measures	DBN	ANN (Moradi et al., 2021)	LSTM	RHUSA+Hybrid Classifier
MAE	0.096	0.100	0.094	0.041
MSE	0.015	0.021	0.014	0.003
RMSE	0.123	0.144	0.118	0.053
MAPE	391.640	109.980	410.040	154.730

Table 9
Classifier performance Evaluation of RHUSA + Hybrid Classifier for M70.

Measures	DBN	ANN (Moradi et al., 2021)	LSTM	RHUSA+Hybrid Classifier
MAE	0.112	0.048	0.102	0.052
MSE	0.017	0.004	0.014	0.004
RMSE	0.130	0.065	0.117	0.061
MAE	398.210	52.755	375.800	158.140

Table 10
Classifier performance Evaluation of RHUSA + Hybrid Classifier for M80.

Measures	DBN	ANN (Moradi et al., 2021)	LSTM	RHUSA+Hybrid Classifier
MAE	0.096	0.036	0.099	0.047
MSE	0.015	0.003	0.016	0.003
RMSE	0.123	0.051	0.127	0.059
MAPE	522.510	72.333	560.090	238.850

Table 11
Classifier performance Evaluation of RHUSA + Hybrid Classifier for M90.

Measures	DBN	ANN (Moradi et al., 2021)	LSTM	RHUSA+Hybrid Classifier
MAE	0.094	0.033	0.100	0.049
MSE	0.014	0.002	0.017	0.004
RMSE	0.120	0.049	0.131	0.062
MAPE	472.690	10.078	569.210	235.320

Table 12
Classifier performance Evaluation of RHUSA + Hybrid Classifier for M100.

Measures	DBN	ANN (Moradi et al., 2021)	LSTM	RHUSA+Hybrid Classifier
MAE	0.096	0.062	0.098	0.051
MSE	0.014	0.007	0.014	0.004
RMSE	0.119	0.084	0.117	0.062
MAPE	23.411	9.853	24.504	11.814

both the M90 and M100 categories. Since the weight of DBN has been optimized with the RHUSA model; the defined objective of error minimization has been achieved. Therefore, the proposed model has stored the minimal error metrics. Among the stored MAE values by the suggested method, the least rate of 0.046649 has been recorded by the proposed work under the M80 category. The adopted model has attained the least MSE value of 0.0027602 under the M60 grade. Hence, from the estimation, it is evident that the suggested method is much more suitable for predicting the concrete strength, as it shows the lowest errors for all 6 grades.

6.6. Performance analysis for dataset 2

According to various algorithms and classifiers, the projected as well as actual output values are depicted in Fig. 6. An evaluation has been conducted to confirm the projected model's ability to accurately estimate compressive strength automatically. According to the findings, there is a strong correlation between the actual and predicted values. It follows that the proposed RHUSA has a lower prediction error and may therefore be used for predicting the strength of concrete with great success.

6.7. Convergence analysis for dataset 2

Fig. 7 shows the convergence analysis for dataset 2. At the 23rd iteration, the suggested method has achieved the minimal cost function of 0.1292. Since the proposed work has succeeded in minimizing errors, it is evident from the evaluation that it is higher recommended for prediction and is hence much more suitable for estimating the strength of the concrete.

6.8. Computational analysis

Table.13 shows the analysis of computational time. The time complexity of the suggested method is low (~344.29) when varying to the traditional models like PSO (~921.76), ABC (~955.6), SS (~873.2), and RH (~373.52). Hence, the efficacy of the suggested method is demonstrated over other conventional methods.

6.9. Ablation analysis

The analysis on ablation study is added in Table 14. The DBN with optimization and without optimization are provided in below table. The MAE, MSE, RMSE, and RAPE are the different measures considered in this research. The error measure of MAE is low for DBN with optimization. Thus, the development of the suggested model is proved.

7. Conclusion

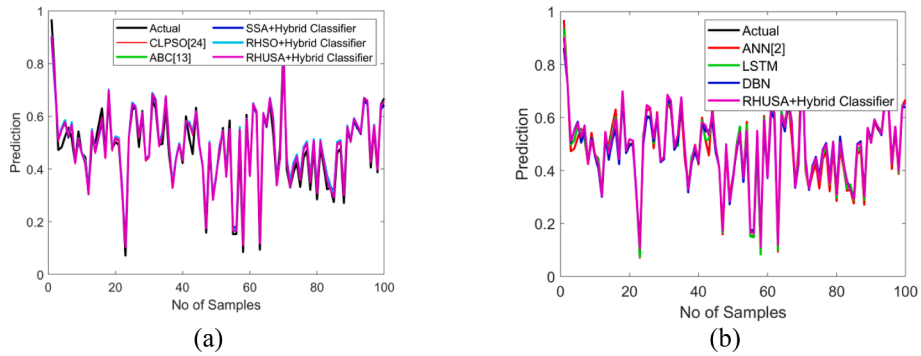
This paper proposed a new model for High-Performance Concrete and of Fiber Reinforced High Strength Self-Compacting Concrete and compression strength prediction is predicted by next three main stages, namely (a) pre-processing, (b) feature extraction, and (c) prediction of concrete compressive strength. Initially, from the collected input data, the replica values were eliminated in the pre-processing stage. Subsequently, multi-features like statistical features (mean, min–max standard deviation, median,), higher order statistical features (variance and skewness), ReliefF feature and symmetrical uncertainty based features were derived from the pre-processed data. To predict the concrete compressive strength phase, a hybrid classifier has been structured by mingling the techniques of DBN and LSTM. Moreover, the weight of DBN has been fine-tuned via RHUSAto improve the prediction accuracy of the suggested work. The proposed hybrid optimization technique is the hybridization of standard SSA and RHSO, respectively. Eventually, the output from the optimized hybrid model will reveal the compressive strength of concrete. Ultimately, a estimation of comparison has been performed to validate the efficiency of the suggested model. On observing the results, the suggested method has attained the minimum value in the case of MAE, MSE, RMSE, and MAPE, respectively. For M50, the MAE of the suggested model is 0.052132, which is 0.052132, which is 23.7%, 15.7%, 15.4%, and 7.1% better than the CLPSO (Li et al., 2019), ABC (keleş et al., 2018), SSA + Hybrid Classifier, RHSO + Hybrid Classifier, respectively.

CRediT authorship contribution statement

Deepa A. Joshi: Conceptualization, Methodology. **Radhika Menon:** Software, Validation. **R.K. Jain:** Investigation, Resources. **A.V. Kulkarni:** Data curation, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.



(a) (b)

Fig. 6. Analysis on performance of dataset 2.

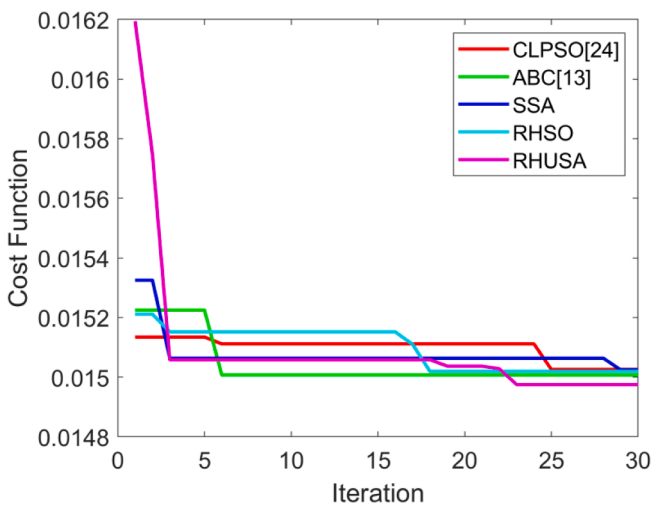


Fig. 7. Convergence Analysis for dataset 2.

Table 13
Analysis on Computational Time.

Methods	Time
PSO	921.76
ABC	955.6
SS	873.2
RH	373.52
RHSS	344.29

Table 14
Analysis on Ablation Study.

Measures	DBN without Optimization	DBN with Optimization
MAE	0.069167	0.043516
MSE	0.0064979	0.002697
RMSE	0.08061	0.051932
RAPE	242.19	132.59

Data availability

The authors do not have permission to share data.

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